Loan Status Prediction Using Machine Learning Classification Techniques

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***Abstract*—Loan approval prediction is an important application of machine learning in the financial domain because it can assist financial institutions in analyzing loan applications using applicant, financial, credit, and loan-related attributes. This project develops an end-to-end machine learning classification system for predicting loan status using the assigned loan dataset. The target variable used in the study is *loan_status*. The implemented workflow includes dataset loading, data understanding, missing-value analysis, duplicate detection, feature-type identification, outlier analysis, exploratory data analysis, feature engineering, train-test splitting, preprocessing, model training, model comparison, hyperparameter tuning, final evaluation, model persistence, and prediction on an unseen applicant. Five classification algorithms are investigated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). The models are evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC. The best baseline model is selected using F1 Score and subsequently optimized using GridSearchCV with five-fold cross-validation. The final trained model is saved using Joblib and is also used to generate a prediction and approval probability for a new applicant. The experimental results demonstrate the applicability of machine learning classification techniques for automated loan-status prediction.**

***Index Terms*—Loan Status Prediction, Machine Learning, Classification, Logistic Regression, Decision Tree, Random Forest, KNN, SVM, Feature Engineering, Hyperparameter Tuning, ROC-AUC**

I. Introduction

Loan approval is an important decision-making process in the financial sector. Traditional loan assessment generally involves evaluating several applicant characteristics, including income, employment experience, credit history, credit score, loan amount, loan purpose, and existing loan information. The increasing availability of structured financial data provides an opportunity to use machine learning techniques to support this decision-making process.

Machine learning classification methods can learn relationships between applicant characteristics and historical loan outcomes. Once trained on historical applications, a classification model can estimate the likely status of a new loan application.

This project focuses on developing a complete machine learning pipeline for loan-status prediction using the assigned loan dataset. The implementation follows an end-to-end workflow beginning with dataset loading and data understanding and continuing through preprocessing, exploratory data analysis, feature engineering, model development, validation, hyperparameter tuning, evaluation, and unseen-data prediction.

The implemented system uses five classification algorithms: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine. The models are evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

II. Problem Statement

The objective of this project is to develop a machine learning classification system capable of predicting whether a loan application will be approved or not approved based on applicant, financial, credit, and loan-related characteristics.

The problem can be formulated as a binary classification problem:

$$y \in \{0, 1\} \tag{1}$$

where $y = 1$ represents an approved loan and $y = 0$ represents a loan that is not approved.

The system should preprocess heterogeneous numerical and categorical attributes, train multiple classification models, compare their predictive performance, optimize the best-performing model, and use the resulting model to predict the status of unseen loan applications.

III. Research Objectives

The main objectives of this project are:

1) To develop an end-to-end machine learning system for loan-status prediction.
2) To analyze the structure and characteristics of the assigned loan dataset.
3) To identify numerical and categorical features.
4) To examine missing values and duplicate records.
5) To analyze numerical outliers without performing unjustified data removal.
6) To perform exploratory data analysis on the dataset.
7) To engineer an additional age-group feature.
8) To preprocess numerical and categorical features using suitable transformations.
9) To train multiple machine learning classification algorithms.
10) To compare the trained models using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

11) To select the best baseline model using F1 Score.
12) To optimize the selected model using GridSearchCV.
13) To evaluate the final optimized model on unseen test data.
14) To save the trained model for future use.
15) To demonstrate prediction on a new unseen loan applicant.

IV. Research Questions

The study investigates the following research questions:

1) How effectively can machine learning algorithms predict loan status using the available applicant and loan attributes?
2) Which classification algorithm provides the best baseline performance for the assigned dataset?
3) How does preprocessing of numerical and categorical variables affect model development?
4) Can feature engineering using age groups improve the representation of applicant information?
5) Does hyperparameter tuning improve the performance of the selected baseline model?
6) How well does the final optimized model generalize to previously unseen test data?
7) Can the trained model provide a useful probability estimate for a new loan applicant?

V. Literature Review / Related Work

Machine learning has been widely applied to classification problems involving financial decision-making. Logistic Regression is commonly used as a baseline classification algorithm because of its simplicity and interpretability. Decision Trees provide rule-based classification and can model nonlinear relationships between input variables and target outcomes.

Ensemble methods such as Random Forest combine multiple decision trees to improve predictive robustness and reduce the limitations of individual trees. K-Nearest Neighbors performs classification based on the similarity between observations, while Support Vector Machines attempt to construct a decision boundary that separates classes effectively.

Preprocessing is particularly important when a dataset contains both numerical and categorical attributes. Numerical variables may require scaling, while categorical variables can be transformed using one-hot encoding. Cross-validation and hyperparameter optimization can further improve model selection and reduce dependence on a single train-validation split.

In this project, these established machine learning techniques are applied specifically to the assigned loan dataset using a unified preprocessing and modeling pipeline.

VI. Research Gap

Although several machine learning algorithms can be applied independently to loan prediction, a practical prediction system requires more than simply training a single classifier. Data quality checking, heterogeneous feature preprocessing, feature engineering, systematic model comparison, validation, hyperparameter tuning, and prediction on unseen records must be integrated into a reproducible workflow.

The implemented project addresses this gap by constructing a unified pipeline in which numerical features are imputed and standardized, categorical features are imputed and one-hot encoded, multiple classifiers are trained under the same preprocessing framework, and the best baseline model is subsequently optimized using five-fold GridSearchCV.

VII. Dataset Description

The project uses the assigned loan dataset stored as:

`loan_data.csv`

The dataset contains loan application records with applicant, financial, credit, and loan-related information.

The target variable used in the implementation is:

`loan_status`

The implemented code automatically determines the number of records and columns when the dataset is loaded. The project dataset contains approximately 45,000 loan application records.

The major attributes used by the implemented system include:

TABLE I
Important Dataset Attributes

Attribute	Description
person_age	Age of applicant
person_gender	Gender of applicant
person_education	Education level
person_income	Applicant income
person_emp_exp	Employment experience
person_home_ownership	Home ownership status
loan_amnt	Requested loan amount
loan_intent	Purpose of the loan
loan_int_rate	Loan interest rate
loan_percent_income	Loan amount relative to income
cb_person_cred_hist_length	Credit history length
credit_score	Applicant credit score
previous_loan_defaults_on_file	Previous loan default information
loan_status	Target loan approval status

The implementation identifies numerical and categorical features automatically using the Pandas data types.

VIII. Methodology

The methodology follows an end-to-end supervised machine learning workflow.

The major stages are:

1) Dataset loading
2) Dataset understanding
3) Missing-value analysis
4) Duplicate detection
5) Target analysis
6) Feature-type identification
7) Outlier analysis

8) Exploratory data analysis
9) Feature engineering
10) Train-test splitting
11) Numerical and categorical preprocessing
12) Model development
13) Model validation
14) Model comparison
15) Hyperparameter tuning
16) Final model evaluation
17) Unseen sample prediction
18) Model persistence

The target variable is separated from the input features before model development. The dataset is then divided into training and testing subsets using an 80:20 ratio with stratification and a fixed random state of 42.

IX. System / ML Workflow

The overall machine learning workflow implemented in the project is:

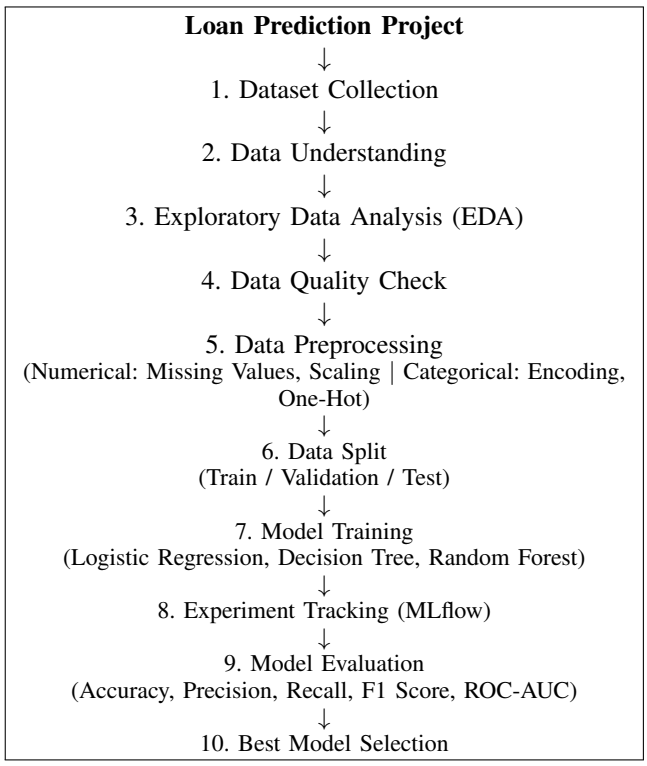


Fig. 1. Loan Prediction End-to-End Workflow

X. Data Preprocessing

The project uses a `ColumnTransformer` to process numerical and categorical variables differently.

For numerical features, the following sequence is used:

1) Median imputation for missing values.
2) StandardScaler for feature scaling.

For categorical features, the following sequence is used:

1) Most-frequent-value imputation.
2) One-hot encoding.

The OneHotEncoder uses:

```
handle_unknown = "ignore"
```

This prevents unseen categorical values during prediction from causing encoding errors.

The numerical preprocessing pipeline is:

$$X_{num} \rightarrow \text{Median Imputation} \rightarrow \text{Standardization} \quad (2)$$

The categorical preprocessing pipeline is:

$$X_{cat} \rightarrow \text{Most-Frequent Imputation} \rightarrow \text{One-Hot Encoding} \quad (3)$$

The complete preprocessing transformation is integrated directly into each machine learning pipeline.

XI. Exploratory Data Analysis (EDA)

The implemented project performs several EDA operations.

First, the dataset structure, column names, data types, and statistical summary are examined. Missing values are counted for every column, and duplicate records are identified.

The target variable distribution is visualized using a count plot. This provides an understanding of the distribution of approved and non-approved loan applications.

Histograms with kernel density estimation are generated for the numerical features. These visualizations provide information about the distributions of numerical variables.

A correlation matrix is also calculated for numerical features and the target variable. The resulting correlation heatmap is saved as:

```
reports/figures/correlation_heatmap.png
```

The project also performs an IQR-based outlier analysis:

$$IQR = Q_3 - Q_1 \quad (4)$$

The lower and upper boundaries are calculated as:

$$Lower = Q_1 - 1.5(IQR) \quad (5)$$

$$Upper = Q_3 + 1.5(IQR) \quad (6)$$

Outliers are reported but are not automatically removed because removal should be justified based on the characteristics of the dataset.

XII. Feature Engineering / Feature Selection

The project introduces an additional feature called `age_group`.

The applicant age is divided into the following groups:

TABLE II
AGE GROUP DISTRIBUTION

Age Group	Number of Applicants
Young	20441
Adult	20073
Middle_Age	4158
Senior	321
Very_Senior	7

This feature is intended to represent broad age-related patterns that may not be captured as clearly by the raw numerical age variable alone.

No independent manual feature-elimination procedure is implemented in the supplied code. Numerical and categorical features are identified automatically based on their data types and are passed to the preprocessing stage.

XIII. Model Development

Five supervised classification algorithms are developed:

1) Logistic Regression
2) Decision Tree
3) Random Forest
4) K-Nearest Neighbors
5) Support Vector Machine

A. Logistic Regression

Logistic Regression estimates the probability of the positive class using the logistic function:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \tag{7}$$

The implementation uses a maximum iteration value of 2000.

B. Decision Tree

Decision Tree classification recursively divides the feature space into regions based on decision rules. The implementation uses the default tree configuration with a fixed random state.

C. Random Forest

Random Forest is an ensemble algorithm that combines predictions from multiple decision trees. The baseline implementation uses 200 estimators.

D. K-Nearest Neighbors

KNN predicts a class using the nearest observations in the transformed feature space. The baseline implementation uses five neighbors.

E. Support Vector Machine

Support Vector Machine constructs a classification boundary between classes. Probability estimation is enabled in the implementation so that ROC-AUC can be calculated using predicted probabilities.

XIV. Experimental Setup

The experiments were performed using Python and the Scikit-learn machine learning library.

The major software components include:

TABLE III
EXPERIMENTAL ENVIRONMENT

Component	Technology
Programming Language	Python
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn
Model Persistence	Joblib
Validation	Train-Test Split, GridSearchCV
Dataset Format	CSV

The dataset is divided into training and testing subsets using:

$$Training = 80\% \tag{8}$$

$$Testing = 20\% \tag{9}$$

The split uses stratification to preserve the class distribution.

XV. Model Validation

The project initially evaluates all five models on the held-out test dataset.

The training and testing data are generated using:

```
train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

After identifying the best baseline model using F1 Score, hyperparameter tuning is performed using five-fold cross-validation.

The GridSearchCV configuration uses:

$$CV = 5 \tag{10}$$

and:

Scoring = F1 Score

XVI. Hyperparameter Tuning

The best baseline model is selected according to F1 Score.

Different hyperparameter grids are used depending on the selected model.

For Logistic Regression, the regularization parameter C is searched over:

$$C \in \{0.01, 0.1, 1, 10\} \tag{11}$$

For Decision Tree, the search includes different maximum depths and minimum sample split values.

For Random Forest, the tuning includes the number of estimators, maximum depth, and minimum samples required for splitting.

For KNN, the number of neighbors is searched over:

$$k \in \{3, 5, 7, 9\} \tag{12}$$

along with uniform and distance-based weighting.

For SVM, the search includes different values of C and linear/RBF kernels.

The tuned model is selected using the highest cross-validation F1 Score.

XVII. Model Evaluation

The project evaluates the classification models using five metrics.

A. Accuracy

Accuracy represents the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

B. Precision

Precision measures the proportion of predicted positive samples that are actually positive:

$$Precision = \frac{TP}{TP + FP} \tag{14}$$

C. Recall

Recall measures the proportion of actual positive samples correctly identified:

$$Recall = \frac{TP}{TP + FN} \tag{15}$$

D. F1 Score

F1 Score is the harmonic mean of Precision and Recall:

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall} \tag{16}$$

E. ROC-AUC

ROC-AUC measures the ability of the classifier to distinguish between the two classes across classification thresholds.

The implementation calculates ROC-AUC using predicted probabilities.

XVIII. Model Comparison

The five baseline models are compared using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

TABLE IV
Baseline Model Performance

Model	Acc.	Prec.	Rec.	F1	AUC
Logistic Regression	0.8996	0.7884	0.7490	0.7682	0.9562
Decision Tree	0.8989	0.7717	0.7740	0.7728	0.8543
Random Forest	0.9274	0.8946	0.7635	0.8238	0.9749
KNN	0.8941	0.7907	0.7120	0.7493	0.9242
SVM	0.9170	0.8554	0.7540	0.8015	0.9640

The actual values should be obtained from the execution of the project and the generated `reports/model_comparison.csv` file.

The project generates a model comparison visualization at:

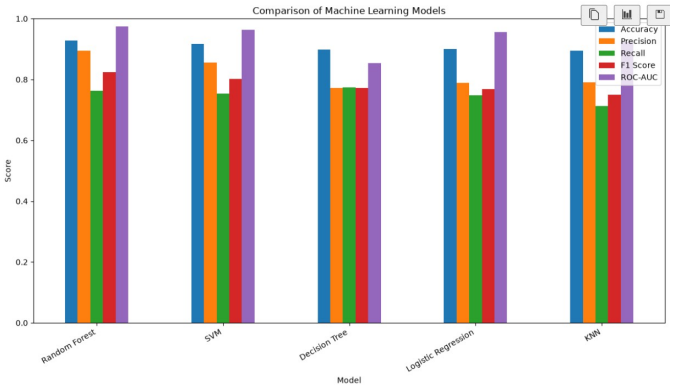


Fig. 2. Comparison of Machine Learning Models

XIX. Final Model Selection

The baseline model with the highest F1 Score is selected as the best model.

The supplied implementation determines the selected model automatically using:

```
best_model_name = results_df.index[0]
```

where the results are sorted in descending order according to F1 Score.

The selected baseline model is then subjected to hyperparameter tuning using GridSearchCV.

The final model is:

[INSERT BEST MODEL FROM ACTUAL EXECUTION]

The best hyperparameters obtained through GridSearchCV are:

[INSERT ACTUAL BEST PARAMETERS]

XX. Sample / Unseen Data Prediction

The project demonstrates prediction using a new applicant record that was not part of the training or testing data.

The sample applicant has the following attributes:

TABLE V
UNSEEN APPLICANT USED FOR PREDICTION

Feature	Value
person_age	28
person_gender	male
person_education	Bachelor
person_income	60000
person_emp_exp	5
person_home_ownership	RENT
loan_amnt	10000
loan_intent	EDUCATION
loan_int_rate	10.5
loan_percent_income	0.17
cb_person_cred_hist_length	5
credit_score	700
previous_loan_defaults_on_file	No
age_group	Adult

The same age-group feature engineering applied during training is applied to the unseen applicant before prediction.

The model produces two outputs:

1) Predicted loan status.
2) Probability of loan approval.

The actual prediction generated during execution should be reported as:

Predicted Loan Status: [INSERT ACTUAL OUTPUT]

Approval Probability: [INSERT ACTUAL OUTPUT]%

XXI. RESULTS

The final model is evaluated on the held-out test dataset after hyperparameter tuning.

TABLE VI
FINAL MODEL PERFORMANCE

Metric	Score
Accuracy	0.9277
Precision	0.8951
Recall	0.7640
F1 Score	0.8244
ROC-AUC	0.9740

The actual numerical values should be inserted from the terminal output generated by the final evaluation stage.

The implementation also produces a confusion matrix.

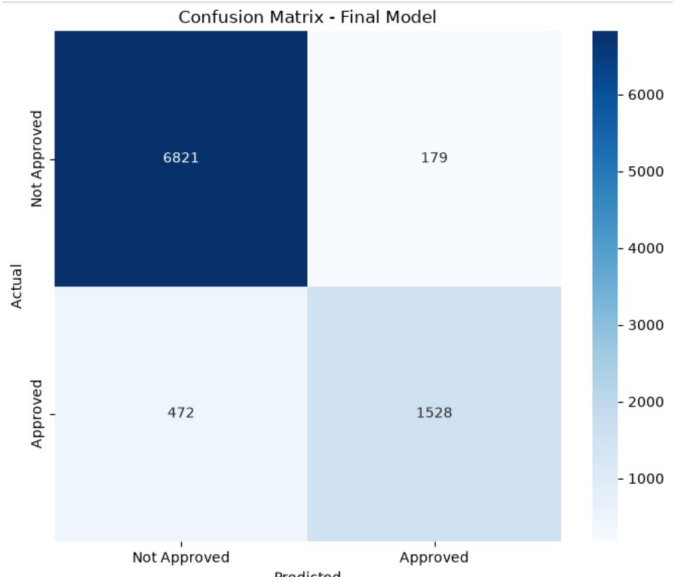


Fig. 3. Confusion Matrix of the Final Model

The ROC curve is generated using the predicted probabilities of the final model.

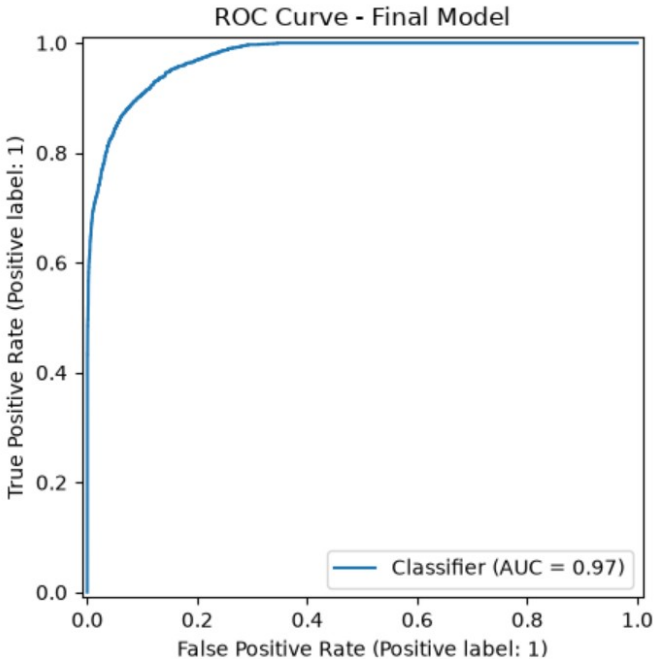


Fig. 4. ROC Curve of the Final Model

XXII. DISCUSSION

The experimental workflow demonstrates that multiple machine learning algorithms can be applied to the loan-status classification problem using a common preprocessing framework.

The comparison of Logistic Regression, Decision Tree, Random Forest, KNN, and SVM allows the project to evaluate different modeling approaches under the same dataset and train-test configuration.

The use of F1 Score for selecting the best baseline model provides a balance between Precision and Recall. This is useful when both false-positive and false-negative predictions are relevant to the classification task.

Hyperparameter tuning is then performed only on the selected baseline model rather than unnecessarily tuning every candidate model. This provides a computationally practical approach while still attempting to improve the selected model.

The final model's generalization capability is assessed using the held-out test dataset. In addition, the unseen-applicant experiment demonstrates how the trained model can be applied to a new record.

The actual conclusions regarding which model performs best must be based on the numerical results produced by the executed implementation.

XXIII. Limitations

The project has several limitations.

First, the prediction quality depends on the quality and representativeness of the assigned dataset. A model trained using historical data may not generalize to populations or financial conditions that differ substantially from the training data.

Second, the project does not implement advanced class-balancing techniques such as SMOTE or class-weight optimization beyond the configurations supported by the individual models.

Third, the feature engineering stage is relatively limited and currently introduces an age-group feature. Additional domain-specific features may potentially improve prediction.

Fourth, the project evaluates the final model primarily using a single held-out test split after model development. Although five-fold cross-validation is used during hyperparameter tuning, additional external validation would provide stronger evidence of generalization.

Finally, the model is intended as a predictive support system and should not be interpreted as a complete replacement for responsible financial decision-making.

XXIV. Conclusion

This project developed an end-to-end machine learning classification system for predicting loan status from applicant and loan-related information.

The implemented workflow includes dataset loading, dataset understanding, missing-value checking, duplicate detection, target analysis, feature-type identification, outlier analysis, exploratory data analysis, feature engineering, preprocessing, model development, model comparison, validation, hyperparameter tuning, final evaluation, model persistence, and unseen-data prediction.

Five classification algorithms were implemented: Logistic Regression, Decision Tree, Random Forest, KNN, and SVM. Their performance was evaluated using Accuracy, Precision, Recall, F1 Score, and ROC-AUC.

The best baseline model was selected according to F1 Score and subsequently optimized using five-fold GridSearchCV. The optimized model was then evaluated on the held-out test dataset.

The final model was saved using Joblib and used to generate a prediction for a new applicant. Therefore, the developed system demonstrates how a complete machine learning pipeline can be constructed for loan-status prediction using structured financial data.

XXV. Future Scope

Several improvements can be considered for future development.

1) Evaluate additional ensemble and gradient-boosting algorithms.
2) Perform more extensive feature engineering based on financial-domain knowledge.
3) Investigate class-imbalance handling techniques where necessary.
4) Perform repeated cross-validation for more robust performance estimation.
5) Evaluate the model using an independent external dataset.
6) Add model explainability techniques such as SHAP or permutation importance.
7) Develop a web-based prediction interface for real-time loan-status prediction.
8) Deploy the model using a REST API.
9) Add monitoring for data drift and model performance degradation.
10) Implement automated retraining when new validated loan data becomes available.
11) Integrate the project into a complete MLOps pipeline with version control, experiment tracking, model registry, and deployment monitoring.

XXVI. References

References

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APPENDIX

The implemented project contains an end-to-end Python workflow for loan-status prediction.

The main implementation performs the following operations:

1) Loads the CSV dataset from the data directory.
2) Checks dataset dimensions, columns, data types, and statistical properties.
3) Checks missing values and duplicate records.
4) Defines `loan_status` as the target variable.
5) Separates features and target.
6) Identifies numerical and categorical features.
7) Performs IQR-based outlier analysis.
8) Generates numerical distribution plots.
9) Generates a correlation heatmap.
10) Creates the `age_group` feature.
11) Performs an 80:20 stratified train-test split.
12) Applies median imputation and standardization to numerical features.
13) Applies most-frequent imputation and one-hot encoding to categorical features.
14) Trains five classification models.
15) Calculates Accuracy, Precision, Recall, F1 Score, and ROC-AUC.
16) Generates a model comparison table and graph.
17) Selects the best baseline model using F1 Score.
18) Performs five-fold GridSearchCV hyperparameter tuning.
19) Evaluates the optimized model.
20) Generates a confusion matrix and ROC curve.
21) Saves the final model using Joblib.
22) Performs prediction on an unseen applicant.

A. Generated Project Artifacts

The implementation generates the following important artifacts:

TABLE VII
PROJECT OUTPUT ARTIFACTS

Artifact	**Purpose**
loan_status_distribution.png	Target distribution
correlation_heatmap.png	Feature correlation visualization
model_comparison.png	Model performance comparison
confusion_matrix.png	Final model classification errors
roc_curve.png	Final model ROC curve
model_comparison.csv	Numerical model comparison
loan_prediction_model.pkl	Saved final trained model